PROJECT DEVELOPMENT PHASE

Global Olympic Sports Performance Analysis

Tableau Dashboards, Data Preparation, Business Questions and Data Storytelling

Project Detail	Information
Prepared by	Aradhana Nagar
Project name	Global Olympic Sports Performance Analysis Using Tableau
Primary tool	Tableau
Dataset	athlete_events.csv (athlete-level Olympic records)
Dataset coverage	1896–2016; Summer and Winter Games
Team ID	To be provided
Submission date	2 November 2026

Prepared using the project details and dataset information available in this conversation. Where an implementation detail or public link was not provided, it is explicitly marked as pending rather than invented.

1. Project Overview

Purpose: This project analyzes historical Olympic athlete and medal records to help users explore country performance, sport-wise medal distribution, athlete-level performance and trends across Olympic editions. Tableau is used to build interactive visualizations, dashboards and a story for decision support and presentation.

Primary users / scenarios:

- **Priya – National Olympic Committee Performance Strategist:** compare country and sport performance, identify sports with strong medal output per athlete, and explore where further training or development analysis may be useful.
- **Carlos – Sports Investment Analyst:** compare medal outcomes and efficiency across countries. Actual investment-versus-medal analysis requires a separate investment dataset; no investment field was present in the supplied athlete_events.csv.
- **William – Olympic Data Storyteller:** communicate medal trends, host-nation comparisons and sport/region patterns. Host-country and region mapping data must be joined from a reliable supplementary source before those analyses can be treated as complete.

Dataset and known limitations

- The supplied athlete_events.csv contains 271,116 rows and 15 columns: ID, Name, Sex, Age, Height, Weight, Team, NOC, Games, Year, Season, City, Sport, Event and Medal.
- The available dataset covers Olympic records from 1896 through 2016. It does not contain 2020 or 2026 records.
- It does not include a Region field, sports investment values or an explicit host-country lookup table.
- Athlete-level data can repeat a team medal across multiple athletes. Medal counting must use a carefully defined event/medal key to avoid overstating team medals.
- Any regional, host-country or investment conclusions remain pending until supplementary data is sourced, joined and validated.

2. Tableau – Business Questions and Visualization Report

The following nine business questions map to the project's three dashboards. The visualization descriptions are implementation targets based on the known project design; insert actual Tableau screenshots and published links after publishing the finished workbook.

No.	Business question	Visualization	Dashboard	Metric / dependency
1	Which sports show the highest average medals per athlete?	Horizontal bar chart: Sport vs. Medals per Athlete; sort descending and show Top 10.	Dashboard 1 – National Olympic Performance	Medals per Athlete = SUM(Medal Flag) / COUNTD(ID). Define Medal Flag as 1 for Bronze, Silver or Gold, otherwise 0.
2	Which NOCs/countries have the highest medal totals?	Ranked bar chart: NOC vs. distinct medal/event count; use a validated unique medal key.	Dashboard 1 – National Olympic Performance	Use COUNTD(Unique Medal Key), not raw row count, to reduce team-medal duplication.
3	How are medals distributed across sports?	Treemap: Sport as label/color and distinct medal count as size; Year and Season filters.	Dashboard 1 – National Olympic Performance	COUNTD(Unique Medal Key), filtered to awarded medals.
4	How does sports investment relate to medal performance by country?	Scatter/bubble plot: investment amount vs. medal count; NOC in detail; bubble size by efficiency.	Dashboard 2 – Investment and Efficiency	Requires a separate country/sport investment dataset; not available in athlete_events.csv.
5	Which regions contribute the most Olympic medals?	Pie/donut chart: Region vs. medal share.	Dashboard 2 – Investment and Efficiency	Requires a validated NOC-to-region mapping table.

No.	Business question	Visualization	Dashboard	Metric / dependency
6	Which countries have higher medal output relative to athlete participation?	Ranked bar chart: NOC vs. medal-per-athlete efficiency; show denominator and selected period.	Dashboard 2 – Investment and Efficiency	Use a clearly labelled project metric; it is not an official Olympic ranking.
7	How have Olympic medal totals changed over time?	Line chart: Year on columns, distinct medal count on rows, Season on color.	Dashboard 3 – Trends and Storytelling	Use validated medal/event key; distinguish Summer and Winter Games.
8	How does the host nation perform in its host edition compared with other editions?	Bar chart comparing host-edition medals with other editions for selected host NOCs.	Dashboard 3 – Trends and Storytelling	Requires a host-city/year/NOC reference table; host status is not directly supplied.
9	How are medals distributed by region and sport?	Donut or stacked bar: Region and Sport with distinct medal count/share.	Dashboard 3 – Trends and Storytelling	Requires NOC-to-region mapping; use filters for Year and Season.

3. Snowflake – Data Preparation Template

Implementation status: The known project workflow uses Tableau with the supplied CSV. No Snowflake database, warehouse, tables or executed SQL outputs have been provided. The following is a reproducible SQL preparation plan for a Snowflake staging layer if the project team chooses to load the same CSV into Snowflake. It must not be reported as already executed.

Suggested data layers

- **RAW:** preserve the imported CSV as received, with source lineage.
- **STAGING:** standardize strings, convert Year/Age/Height/Weight to numeric types where valid, and normalize Medal values.
- **CURATED:** publish validated athlete and medal/event views for analysis.
- **MART:** create aggregate views for country, sport, year and season dashboards.

Activity 1 – Identify and handle NULL values

Missing Age, Height and Weight may be retained as NULL because they are not required for every medal analysis. Do not replace missing measurements with zero. Medal NULL means no medal was recorded and should not be counted as an award. Check missingness before choosing a treatment.

```
-- Example after loading the CSV into RAW.ATHLETE_EVENTS
SELECT
COUNT(*) AS total_rows,
COUNT_IF(ID IS NULL) AS missing_id,
COUNT_IF(NOC IS NULL) AS missing_noc,
COUNT_IF(Year IS NULL) AS missing_year,
COUNT_IF(Sport IS NULL) AS missing_sport,
COUNT_IF(Medal IS NULL OR UPPER(Medal) = 'NA') AS no_medal_value
FROM RAW.ATHLETE_EVENTS;

-- Create a staging view; retain legitimate unknown measurements as NULL.
CREATE OR REPLACE VIEW STAGING.ATHLETE_EVENTS_CLEAN AS
SELECT
ID, TRIM(Name) AS Name, UPPER(TRIM(Sex)) AS Sex,
TRY_TO_NUMBER(Age) AS Age,
TRY_TO_NUMBER(Height) AS Height,
TRY_TO_NUMBER(Weight) AS Weight,
TRIM(Team) AS Team, UPPER(TRIM(NOC)) AS NOC,
TRIM(Games) AS Games, TRY_TO_NUMBER(Year) AS Year,
TRIM(Season) AS Season, TRIM(City) AS City,
TRIM(Sport) AS Sport, TRIM(Event) AS Event,
CASE WHEN UPPER(TRIM(Medal)) IN ('GOLD','SILVER','BRONZE')
THEN INITCAP(TRIM(Medal)) ELSE NULL END AS Medal
FROM RAW.ATHLETE_EVENTS;
```

Expected validation: the first query reports null counts; the staging view preserves missing measurements as NULL and maps non-medal markers to NULL. Actual output values depend on the Snowflake-loaded data and are not fabricated here.

Activity 2 – Duplicate record handling

Repeated athlete-event rows may be valid when they represent different events or Games. Do not delete records solely because names repeat. Use a composite key to inspect exact duplicates, then retain one copy only when all identifying fields match.

```
SELECT ID, Games, Event, NOC, Medal, COUNT(*) AS duplicate_count
FROM STAGING.ATHLETE_EVENTS_CLEAN
GROUP BY ID, Games, Event, NOC, Medal
HAVING COUNT(*) > 1
ORDER BY duplicate_count DESC;

-- Example deduplicated view; verify the chosen key against source semantics.
CREATE OR REPLACE VIEW CURATED.ATHLETE_EVENTS AS
SELECT *
FROM STAGING.ATHLETE_EVENTS_CLEAN
QUALIFY ROW_NUMBER() OVER (
PARTITION BY ID, Games, Event, NOC, Medal
ORDER BY ID
) = 1;
```

Expected validation: duplicate groups are listed for review; the curated view retains one row per chosen composite key. Review team-event records and ties before using this view for official medal totals.

Activity 3 – Data cleaning and standardization

```
-- Check categorical values and invalid year range.
SELECT DISTINCT Sex FROM STAGING.ATHLETE_EVENTS_CLEAN;
SELECT DISTINCT Season FROM STAGING.ATHLETE_EVENTS_CLEAN;
SELECT DISTINCT Medal FROM STAGING.ATHLETE_EVENTS_CLEAN;
SELECT MIN(Year) AS first_year, MAX(Year) AS last_year
FROM STAGING.ATHLETE_EVENTS_CLEAN;

-- Standardized curated output.
CREATE OR REPLACE VIEW CURATED.ATHLETE_EVENTS_STANDARD AS
SELECT
ID, Name,
CASE WHEN Sex IN ('M','F') THEN Sex ELSE NULL END AS Sex,
Age, Height, Weight, Team, NOC, Games, Year,
INITCAP(Season) AS Season, City, Sport, Event, Medal
FROM CURATED.ATHLETE_EVENTS
WHERE Year BETWEEN 1896 AND 2016
AND NOC IS NOT NULL
AND Sport IS NOT NULL;
```

Expected validation: category checks expose unexpected values; year bounds reflect the supplied dataset coverage. Do not silently discard unusual but legitimate records without documenting the rule.

Activity 4 – Transformation and validation

```
CREATE OR REPLACE VIEW MART.SPORT_MEDAL_SUMMARY AS
SELECT
Sport, Season, Year,
COUNT(DISTINCT ID) AS athlete_count,
COUNT_IF(Medal IS NOT NULL) AS medal_awarded_rows
FROM CURATED.ATHLETE_EVENTS_STANDARD
GROUP BY Sport, Season, Year;

-- Medal records by country/NOC for exploratory reporting.
CREATE OR REPLACE VIEW MART.NOC_MEDAL_ROWS AS
SELECT NOC, Year, Season, Medal, Sport, Event, ID
FROM CURATED.ATHLETE_EVENTS_STANDARD
WHERE Medal IN ('Gold','Silver','Bronze');

-- Basic validation checks
SELECT COUNT(*) AS curated_rows FROM CURATED.ATHLETE_EVENTS_STANDARD;
SELECT COUNT_IF(Medal IS NOT NULL) AS medal_rows
FROM CURATED.ATHLETE_EVENTS_STANDARD;
SELECT COUNT(DISTINCT NOC) AS noc_count
FROM CURATED.ATHLETE_EVENTS_STANDARD;
```

Important: medal_awarded_rows is a row-level measure, not necessarily the official number of medals. For team events, create and validate a unique event-medal identifier using event, Games, NOC and medal/tie logic before using COUNT(DISTINCT key) as the medal count.

4. Snowflake Project – Code and Solution Summary

Project / repository link: Not provided. Add a GitHub/GitLab link only if a repository is created.

Data source: User-provided athlete_events.csv. The file contains 271,116 rows and 15 columns, covering 1896–2016. Supplementary region, host and investment datasets are not yet evidenced.

Database / warehouse: Not created or confirmed. Proposed names if Snowflake is adopted: OLYMPIC_ANALYTICS database; OLYMPIC_WH warehouse. These are suggestions, not existing resources.

Tables/views: Proposed RAW.ATHLETE_EVENTS, STAGING.ATHLETE_EVENTS_CLEAN, CURATED.ATHLETE_EVENTS_STANDARD, MART.SPORT_MEDAL_SUMMARY and MART.NOC_MEDAL_ROWS.

Key transformations: trim and standardize text; convert numeric columns safely; normalize Medal values; preserve unknown measurements as NULL; inspect duplicates with a composite key; derive medal flags and athlete/medal aggregates; validate year and category ranges.

KPIs: athlete count, medal-awarded athlete rows, distinct NOC count, medals by sport, medals by NOC, average medals per athlete, medal trend by year/season. Investment return and regional share require extra datasets.

Key insights: The dashboard is designed to identify sports with high medal output per athlete, compare NOC performance, show sport-wise medal distribution and track trends over time. Actual ranked findings should be taken from the finished Tableau workbook, not assumed from example figures.

Quality checklist

Criterion	Status	Evidence / next action
Data loaded into Snowflake	Pending	CSV is available; Snowflake load not confirmed.
Data cleaned and transformed	Partially planned	Tableau preparation / SQL examples documented; execution evidence needed.
Tables and relationships implemented	Pending	Create and validate proposed layers if Snowflake is required.
SQL calculations and KPIs implemented	Pending	Run scripts and compare outputs with Tableau.
Outputs validated	Pending	Capture row counts, null checks, duplicate checks and reconciliation.
Solution documented	Yes – documentation draft	This report records the design and clearly labels unexecuted steps.

5. Tableau – Dashboard Design

Dashboard 1: National Olympic Performance

- Visual 1: Average Medals per Athlete by Sport – horizontal bar chart; Top 10 sports; descending sort.
- Visual 2: Global Performance by Country/NOC – ranked bar chart using a validated medal key.
- Visual 3: Medal Distribution by Sport – treemap with Sport as label/color and medal count as size.
- Purpose: help the NOC strategist compare sports and country performance.

Dashboard 2: Sports Investment and Regional Efficiency

- Visual 4: Sports Investment vs Medal Performance – scatter/bubble chart. Requires an investment dataset and matching country/sport identifiers.

- Visual 5: Regional Medal Efficiency – pie/donut chart. Requires a verified NOC-to-region lookup.
- Visual 6: Country-wise Medal Efficiency – ranked bar chart with a clearly defined denominator and selected time period.
- Purpose: support resource-allocation discussion while avoiding claims that investment causes medal outcomes.

Dashboard 3: Olympic Trends and Storytelling

- Visual 7: Medal Trends by Year and Season – line chart with Summer/Winter color distinction.
- Visual 8: Host Nation Medal Comparison – bar chart using a host-city/year lookup joined to NOC results.
- Visual 9: Regional Medal Distribution by Sport – donut or stacked bar using region mapping.
- Purpose: explain long-term trends, host-edition comparisons and geographic patterns.

Design and interaction standards

- Use a light background, dark navy titles and sky-blue accents; use gold, silver/grey and bronze/brown consistently for medal categories.
- Keep titles descriptive, legends readable, axes labelled and number formats consistent.
- Add Year, Season, NOC, Sport and Medal filters where relevant; use dropdown or multi-value controls without overcrowding.
- Use dashboard actions/highlighting to let a selection in one chart filter or highlight related charts.
- Use a consistent grid, aligned containers, balanced whitespace and readable sizing; test on the intended screen size.
- Add tooltips explaining the metric, data coverage and limitations. Avoid unsupported Region/Investment fields until data is joined.

Major expected outcomes

- Identify sports with relatively high medal output per athlete.
- Compare medal performance across NOCs and sports.
- Explore how medal distributions vary across sports.
- Trace medal trends across Olympic years and seasons.
- Provide a framework for investment and regional analysis once the missing data sources are added.

Tableau Public publication

Public dashboard URL: Not provided. Paste the published Tableau Public dashboard links here after publishing and checking that each view opens without sign-in.

Screenshots: Not embedded because no finished dashboard screenshot files were supplied in this upload. Insert screenshots of Dashboard 1, Dashboard 2 and Dashboard 3 in the final institutional submission if required.

6. Snowflake / Streamlit Dashboard Design Template – Applicability

The uploaded template describes a Streamlit dashboard backed by Snowflake. The project currently documented here is a Tableau dashboard project; no Streamlit application or Snowflake connection has been confirmed. To keep the submission accurate, this section records the equivalent dashboard design and marks Streamlit implementation as pending.

- Clear layout: organize the page into overview KPIs, country/sport comparisons and trends.
- Appropriate visuals: use bar charts, treemaps, line charts, scatter plots and regional charts only when the required data exists.
- Theme: light background with navy and sky-blue accents, matching Tableau.

- Filters: Year, Season, NOC, Sport and Medal; Region only after mapping is added.
- Drill-down: select a country to inspect sports, then select a sport to inspect yearly performance.
- Responsive design: use containers/columns and test desktop and smaller screens.
- Custom visuals/infographics: medal icons and concise KPI cards may be used with clear labels.

Streamlit app URL: Not created / not provided. Do not present this as published until an app is implemented and deployed.

7. Tableau Story Design

The Tableau Story is planned as four story points that guide the audience from context to the three analysis scenarios.

Story point	Title	Narrative / content
1	Olympic Games – Project Overview	Introduce the project objective, dataset coverage (1896–2016), Summer/Winter scope, key fields and known limitations.
2	Scenario 1 – National Olympic Performance	Use average medals per athlete, NOC medal ranking and sport-wise medal distribution to compare performance.
3	Scenario 2 – Sports Investment and Efficiency	Explain country efficiency and investment-versus-medal comparison. Clearly state that investment and region data must be added before these views are fully supported.
4	Scenario 3 – Olympic Trends and Storytelling	Present medal trends by year/season, host-edition comparisons and regional/sport patterns where supplementary mappings are available.

Story observations / key messages

- Sport-wise averages can highlight disciplines worth investigating, but averages should be interpreted alongside participation counts and event structure.
- Country medal totals provide a broad comparison but are influenced by delegation size, number of events and historical participation.
- Year and season filters help reveal changes across Summer and Winter Games.
- Investment efficiency cannot be concluded without actual investment data and a defensible time period.
- Host-nation and regional findings require verified reference data and should not be inferred from city or NOC labels alone.

Tableau Public Story URL: Not provided. Add the published Story URL after publishing and validating all four story points.

8. Development Plan, Deliverables and Validation

Work package	Planned work / deliverable	Status based on known project information
Data intake	Use athlete_events.csv; document source, columns, coverage and limitations.	Dataset supplied and inspected.
Data preparation	Clean fields, inspect nulls/duplicates, define medal counting logic.	Preparation approach documented; exact Tableau workbook calculations need final verification.
Visualization build	Create nine visualizations across three scenarios.	Design and worksheet plan documented; verify final workbook.
Dashboard build	Assemble three dashboards with filters, actions and consistent theme.	Dashboard work reported by user; final screenshots/links not supplied here.

Work package	Planned work / deliverable	Status based on known project information
Story build	Create four story points: overview + three scenarios.	User previously reported four story points created.
Supplementary data	Add NOC-region, host-edition and investment data where needed.	Pending; not in supplied CSV.
Testing	Check filter behaviour, totals, duplicate handling, labels, layout and load time.	Testing evidence pending.
Publication	Publish Tableau dashboards and story; record public URLs.	Links not provided.
Documentation	Capture screenshots, metric definitions, source notes and limitations.	This PDF provides the development documentation draft.

Risks and mitigation

- Team medal duplication: validate event-level medal keys and compare totals with a trusted reference before presenting official counts.
- Missing region/investment/host fields: obtain reliable supplementary sources, document join keys and test unmatched NOCs.
- Different participation levels: show athlete/event denominators alongside medal counts.
- Incomplete time coverage: label the data range as 1896–2016 and do not claim 2020–2026 coverage.
- Dashboard clutter or slow performance: limit default views, use relevant filters and test workbook responsiveness.

Final deliverables checklist

- Three Tableau dashboards (National Olympic Performance; Investment and Efficiency; Trends and Storytelling).
- Nine business questions with corresponding visualizations.
- Four-point Tableau Story.
- Data preparation and validation documentation.
- SQL examples for an optional Snowflake staging workflow.
- Published Tableau Public URLs and screenshots – to be inserted when available.
- Supplementary datasets for region, host and investment analysis – pending.

9. Assumptions and Items Requiring Confirmation

- Team ID was not available; it is intentionally left as “To be provided”.
- No Snowflake database/warehouse, Streamlit app, Git repository or public Tableau links were supplied.
- The report does not invent chart values, project results, screenshots or publication URLs.
- The supplied dataset's known coverage is 1896–2016; references in earlier project drafts to 2020 or 2026 should be corrected unless newer data is added.
- Before final submission, replace pending statuses with actual evidence from the workbook/platform and attach screenshots where the college template asks for them.